

INTRODUCTION TO MODERN CRYPTOGRAPHY

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Part I

Private-key Cryptography

Chapter 1

Introduction

Let $\mathcal{K}, \mathcal{M}, \mathcal{C}$ be finite sets.

1.1 Notations and Definitions

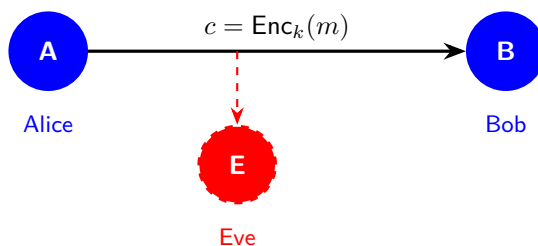
Definition 1.1 [Encryption] A function $\text{Enc} : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{C}$. For all $k \in \mathcal{K}, m \in \mathcal{M}$, the message after the encryption, $c = \text{Enc}(k, m) \triangleq \text{Enc}_k(m)$, is the **ciphertext**.

Definition 1.2 [Decryption] A function $\text{Dec} : \mathcal{K} \times \mathcal{C} \rightarrow \mathcal{M}$. For all $k \in \mathcal{K}, c \in \mathcal{C}$, the message before the encryption, $m = \text{Dec}(k, c) \triangleq \text{Dec}_k(c)$, is the **plaintext**.

Definition 1.3 [Correctness] For all $m \in \mathcal{M}$, $\text{Dec}_k(\text{Enc}_k(m)) = m$.

1.1.1 Communication Model

Let us welcome the three major players in this field, **Alice**, **Bob**, and **Eve** (claps!).



In our model, the "major players" interact as follows:

- **Parties:** Alice and Bob are the legitimate participants.
- **Adversary:** Eve is the eavesdropper monitoring the line.
- **Channel:** A reliable communication line exists between Alice and Bob.
- **The Shared Secret:** They utilize an encryption scheme (Enc, Dec) that is *known* to Eve, and a *secret* key k .
- **The Goal:** Alice wants to send a message m to Bob such that it remains *confidential*.

1.2 Perfect Secrecy

Definition 1.4 [Perfect Secrecy (Shannon 1949)] Two distributions X, Y over $\mathcal{D}$ are **equal**, denoted by $X \equiv Y$, if $\forall d \in \mathcal{D}. \Pr[X = d] = \Pr[Y = d]$.

We say (Enc, Dec) is **perfectly secret** if for any a priori distribution M of messages, and $k \xleftarrow{R} \mathcal{K}$, $M|\text{Enc}_k(M) \equiv M$.

Definition 1.5 [Perfect Indistinguishability] We say (Enc, Dec) is **perfectly indistinguishable** if for any $m_0, m_1 \in \mathcal{M}$ of equal length, if $k \xleftarrow{R} \mathcal{K}$, then $\text{Enc}_k(m_0) \equiv \text{Enc}_k(m_1)$.

Definition 1.6 [Indistinguishability Game] For an *adversary* $\mathcal{A} : \mathcal{C} \rightarrow \{0, 1\}$ we define the following game:

- $\mathcal{A}$ chooses messages $m_0, m_1 \in \mathcal{M}$.
- We sample a key $k \xleftarrow{R} \mathcal{K}$ and a bit $b \leftarrow \{0, 1\}$.
- We send $\text{Enc}_k(m_b)$ to $\mathcal{A}$, called the *challenge*.
- $\mathcal{A}$ outputs a bit b' .

We say that $\mathcal{A}$ **wins** if $b' = b$, and we denote $\Pr[\mathcal{A} \text{ wins}] \triangleq \Pr[b = b']$.

Theorem 1.1 Let (Enc, Dec) be a symmetric encryption scheme. Then, the following are equivalent:

1. (Enc, Dec) is *perfectly secret*.
2. (Enc, Dec) is *perfectly indistinguishable*.
3. For any adversary $\mathcal{A}$, $\Pr[\mathcal{A} \text{ wins}] = \frac{1}{2}$.

Proof

(1) $\iff$ (2) was left to HW1, Q1. (2) $\implies$ (3): Let $\mathcal{A}$ be an adversary, w.l.o.g deterministic, and let $m_0, m_1 \in \mathcal{M}$. Define $S_b \triangleq \mathcal{A}^{-1}(\{b\})$ for $b \in \{0, 1\}$, then $S_0 \uplus S_1 = \mathcal{C}$. Let $b' = \mathcal{A}(m_b)$, where $b \leftarrow \{0, 1\}$, since (Enc, Dec) is p.i., then by total probability

$$\begin{aligned} \Pr[\mathcal{A} \text{ wins}] &= \Pr[b' = b | b = 0] \Pr[b = 0] + \Pr[b' = b | b = 1] \Pr[b = 1] \\ &= \frac{1}{2} (\Pr[\text{Enc}_k(m_0) \in S_0] + \Pr[\text{Enc}_k(m_1) \in S_1]) \\ &= \frac{1}{2} (\Pr[\text{Enc}_k(m_0) \in S_0] + \Pr[\text{Enc}_k(m_0) \in S_1]) \\ &= \frac{1}{2} \Pr[\text{Enc}_k(m_0) \in S_0 \uplus S_1] = \frac{1}{2}. \end{aligned}$$

(3) $\implies$ (2): Suppose (Enc, Dec) is not p.i., then there must be some $m_0, m_1 \in \mathcal{M}, c^* \in \mathcal{C}$ s.t.

$$\alpha = \Pr[\text{Enc}_k(m_0) = c^*] > \Pr[\text{Enc}_k(m_1) = c^*] = \beta.$$

Consider the adversary $\mathcal{A}$ that selects the messages m_0, m_1 , and on input $c \in \mathcal{C}$, $\mathcal{A}(c) = \mathbf{1}[c \neq c^*]$. Then by total probability

$$\begin{aligned} \Pr[\mathcal{A} \text{ wins}] &= \Pr[b' = b | b = 0] \Pr[b = 0] + \Pr[b' = b | b = 1] \Pr[b = 1] \\ &= \frac{1}{2} (\Pr[\text{Enc}_k(m_0) = c^*] + \Pr[\text{Enc}_k(m_1) \neq c^*]) \\ &= \frac{1}{2} (\alpha + (1 - \beta)) = \frac{1}{2} + \frac{1}{2} (\alpha - \beta) > \frac{1}{2}. \end{aligned}$$

■

1.3 One-Time Pad

Definition 1.7 [One-Time Pad] Let $\mathcal{M} = \mathcal{K} = \mathcal{C} = \{0, 1\}^n$, then for any $m, k, c \in \{0, 1\}^n$ (where $\oplus$ is bitwise XOR), we define the **one-time pad** scheme:

$$\text{Enc}_k(m) = m \oplus k,$$

$$\text{Dec}_k(c) = c \oplus k.$$

Theorem 1.2 One-Time Pad is *perfectly secret*.

Proof

It is suffice to show the following claim:

Claim 1.1 For any $m, c \in \{0, 1\}^n$

$$\Pr_{k \leftarrow \{0, 1\}^n} [\text{Enc}_k(m) = c] = \frac{1}{2^n} \iff \text{Enc}_k(m) \sim \mathcal{U}_n$$

The claim implies the theorem from transitivity, for any $m_0, m_1 \in \{0, 1\}^n$, where $k \xleftarrow{R} \{0, 1\}^n$:

$$\text{Enc}_k(m_0) \equiv \mathcal{U}_n \equiv \text{Enc}_k(m_1)$$

Proof of the claim. Let $m, c \in \{0, 1\}^n$, then:

$$\Pr_k[\text{Enc}_k(m) = c] = \Pr_k[m \oplus k = c] = \Pr_k[k = m \oplus c] = \frac{1}{2^n}$$

since $m \oplus c$ is a constant and $k \sim \mathcal{U}_n$. ■

1.4 Shannon's Theorem

Theorem 1.3 [Shannon] If (Enc, Dec) is perfectly secret, then $|\mathcal{K}| \geq |\mathcal{M}|$.

Proof

Define a two-sided graph $G = (V, E)$ where $V = \mathcal{M} \uplus \mathcal{C}$, $\{m, c\} \in E$ if and only if there exists $k \in \mathcal{K}$ s.t. $\text{Enc}_k(m) = c$. Assume for sake of contradiction that $|\mathcal{K}| < |\mathcal{M}|$, and let $\{m, c\} \in E$.

Claim 1.2 c is connected to at most $|\mathcal{K}|$ messages, i.e. $\deg(c) \leq |\mathcal{K}|$.

Proof of the claim. Assume for sake of contradiction that $\deg(c) > |\mathcal{K}|$. By the pidgeonhole principle, there exists a key $k \in \mathcal{K}$ and two messages $m_0, m_1 \in \mathcal{M}$ s.t. $c = \text{Enc}_k(m_0) = \text{Enc}_k(m_1)$ and that is a contradiction to the correctness of (Enc, Dec) .

Corollary 1.1 There exists a message $m^* \in \mathcal{M}$ that isn't a neighbor of c .

For any key $k \in \mathcal{K}$, it holds that $\text{Enc}_k(m^*) \neq c$, and thus $\Pr_k[\text{Enc}_k(m^*) = c] = 0$.

Since $\{m, c\} \in E$, there exists a key $k' \in \mathcal{K}$ s.t. $\text{Enc}_{k'}(m) = c$, and thus $\Pr_k[\text{Enc}_k(m) = c] \geq \frac{1}{|\mathcal{K}|} > 0$.

We then conclude $\text{Enc}_k(m^*) \neq \text{Enc}_k(m)$, where $k \leftarrow \mathcal{K}$, which is a contradiction. ■

Chapter 2

Computational Secrecy & Pseudorandomness

2.1 Computational Secrecy

Claim 2.1 [Perfect Secrecy Revisited] (Enc, Dec) has perfect secrecy if and only if for any $m_0, m_1 \in \mathcal{M}$ of equal length, if we denote $c_0 = \text{Enc}_k(m_0), c_1 = \text{Enc}_k(m_1)$ for all $k \in \mathcal{K}$, then for any adversary $\mathcal{A}$,

$$\Pr_{k \leftarrow^R \mathcal{K}} [\mathcal{A}(c_0) = 1] = \Pr_{k \leftarrow^R \mathcal{K}} [\mathcal{A}(c_1) = 1]$$

Definition 2.1 [Adversary Complexity] We say $\mathcal{A}$ is of **complexity** t , if it's computable by a circuit of size t .

Definition 2.2 [Distinguishing Advantage] Let X, Y be two distributions over $\mathcal{D}$, and let $\mathcal{A} : \mathcal{D} \rightarrow \{0, 1\}$ be an adversary, we define $\mathcal{A}$'s **distinguishing advantage** as

$$\Delta_{\mathcal{A}}(X, Y) \triangleq |\Pr[\mathcal{A}(X) = 1] - \Pr[\mathcal{A}(Y) = 1]|$$

Definition 2.3 $[(t, \varepsilon)$ -indistinguishability] We say X and Y are (t, ε) -**indistinguishable** and denote $X \approx_{t, \varepsilon} Y$, if for all adversaries $\mathcal{A}$ of complexity at most t , $\Delta_{\mathcal{A}}(X, Y) \leq \varepsilon$.

Remark 2.1 $X \equiv Y \iff X \approx_{\infty, 0} Y$.

Claim 2.2 [Properties of Indistinguishability] Let X, Y, Z be distributions over $\mathcal{D}$, $\varepsilon, \varepsilon_1, \varepsilon_2, t$ and $f : \mathcal{D} \rightarrow \mathcal{D}$.

1. If $X \approx_{t, \varepsilon} Y$ and f is of complexity at most ℓ , then $f(X) \approx_{t-\ell, \varepsilon} f(Y)$.
2. If $X \approx_{t, \varepsilon_1} Y \approx_{t, \varepsilon_2} Z$ then $X \approx_{t, \varepsilon_1 + \varepsilon_2} Z$.

Definition 2.4 [Computational Secrecy (Goldwasser & Micali '82)] We say (Enc, Dec) is (t, ε) -**computationally secret** if for any $m_0, m_1 \in \mathcal{M}$ of equal length, if we denote $c_0 = \text{Enc}_k(m_0), c_1 = \text{Enc}_k(m_1)$ for any $k \in \mathcal{K}$, then for any adversary $\mathcal{A}$ of complexity at most t :

$$\Pr_{k \leftarrow^R \mathcal{K}} [\mathcal{A}(c_0) = 1] - \Pr_{k \leftarrow^R \mathcal{K}} [\mathcal{A}(c_1) = 1] < \varepsilon,$$

i.e. $c_0 \approx_{t, \varepsilon} c_1$.

Remark 2.2 Perfect secrecy is $(\infty, 0)$ -computational secrecy.

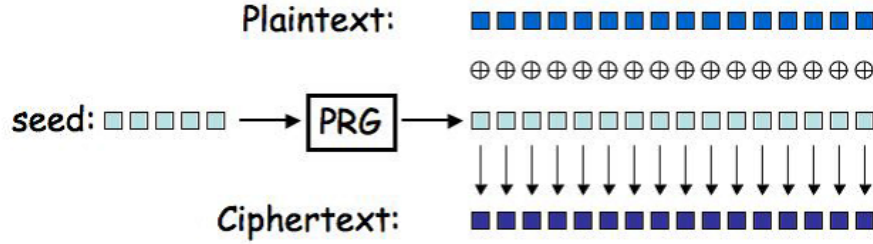
2.2 Pseudo Random Generators

Definition 2.5 [PRG] A (t, ε) -pseudo random generator is a polynomial time computable function $G : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$, $\ell \gg n$, if we denote $y_0 = G(k)$ for any $k \in \{0, 1\}^n$, that for any adversary $\mathcal{A}$ of complexity at most t :

$$\Pr_{k \leftarrow^R \{0, 1\}^n} [\mathcal{A}(y_0) = 1] - \Pr_{y_1 \leftarrow^R \{0, 1\}^\ell} [\mathcal{A}(y_1) = 1] < \varepsilon,$$

i.e. $y_0 \approx_{t, \varepsilon} y_1$.

2.2.1 Computational One-Time Pad



Definition 2.6 [Computational One-Time Pad] Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$, $\ell \gg n$ be a PRG, then we define a scheme (Enc, Dec) for $\mathcal{K} = \{0, 1\}^n$, $\mathcal{M} = \mathcal{C} = \{0, 1\}^\ell$:

- Choose a secret random short key $k \leftarrow^R \{0, 1\}^n$ ("seed")
- Expand the seed into a long *keying stream* $G(k)$
- Encrypt m by $\text{Enc}_k(m) = G(k) \oplus m$
- Decrypt c by $\text{Dec}_k(c) = c \oplus G(k)$

Theorem 2.1 Assume that $\text{PRG} : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$ is (t, ε) -pseudorandom. Then the "computational OTP" is $(t - \ell, 2\varepsilon)$ -secure.

Proof

Let $m_0, m_1 \in \mathcal{M}$, we need to show $(\text{PRG}(\mathcal{U}_n) \oplus m_0) \approx_{t', \varepsilon'} (\text{PRG}(\mathcal{U}_n) \oplus m_1)$, where $t' = t - \ell$, $\varepsilon' = 2\varepsilon$.

Claim 2.3 $(\text{PRG}(\mathcal{U}_n) \oplus m) \approx_{t', \varepsilon} (\mathcal{U}_\ell \oplus m) \equiv \mathcal{U}_\ell$

From transitivity we conclude $(\text{PRG}(\mathcal{U}_n) \oplus m_0) \approx_{t', \varepsilon'} (\text{PRG}(\mathcal{U}_n) \oplus m_1)$.

Proof of the claim. Assume for sake of contradiction there exists an adversary $\mathcal{A}$ of complexity t' such that $\Delta_{\mathcal{A}}(\text{PRG}(\mathcal{U}_n) \oplus m_0, \mathcal{U}_\ell \oplus m) > \varepsilon$, we will describe an adversary breaking the PRG.

$\mathcal{A}'(z) = \mathcal{A}(z \oplus m)$, and it's easy to see that $\Delta_{\mathcal{A}'}(\text{PRG}(\mathcal{U}_n), \mathcal{U}_\ell) = \Delta_{\mathcal{A}}(\text{PRG}(\mathcal{U}_n) \oplus m, \mathcal{U}_\ell \oplus m) > \varepsilon$.

The complexity of $\mathcal{A}'$ is at most $t' + \ell = t$ and thus we've reached a contradiction. ■

Remark 2.3 PRGs are central primitives in theory of cryptography. If $\mathbf{P} = \mathbf{NP}$ then there are no (polynomial) PRGs, $L = \{z : \exists x. \text{PRG}(x) = z\} \in \mathbf{NP}$, so the existence of PRGs can only be conjectured until $\mathbf{P}$ vs $\mathbf{NP}$ is resolved, but we have good reasons to believe they do exist. However there are good reasons to believe that PRGs do exist. Many candidates are known whose security can be reduced to seemingly hard problems. Furthermore, it was shown that PRGs can be constructed based on any *One-way function*.

2.3 Multiple Messages

Definition 2.7 [Computational Secrecy for Multiple Messages] We say (Enc, Dec) is (t, ε) -**computationally secret for multiple messages** if for any $\vec{x} = (x_1, \dots, x_n), \vec{y} = (y_1, \dots, y_n) \in \mathcal{M}^n$, s.t. $\{(x_i, y_i)\}_{i=1}^n$ are pairs of equal length messages, if we denote $\vec{c}_0 = (\text{Enc}_k(x_1), \dots, \text{Enc}_k(x_n)), c_1 = (\text{Enc}_k(y_1), \dots, \text{Enc}_k(y_n))$ for any $k \in \mathcal{K}$, then for any adversary $\mathcal{A}$ of complexity at most t :

$$\Pr_{k \leftarrow \mathcal{K}} [\mathcal{A}(\vec{c}_0) = 1] - \Pr_{k \leftarrow \mathcal{K}} [\mathcal{A}(\vec{c}_1) = 1] < \varepsilon,$$

i.e. $\vec{c}_0 \approx_{t, \varepsilon} \vec{c}_1$.

2.3.1 Insecurity

Theorem 2.2 If (Enc, Dec) is deterministic and dependent on the key and message k, m only, then it is *insecure for multiple messages*.

Proof

Let $a, b \in \mathcal{M}$ be two different messages of the same length. Then for

$$\vec{x} = (a, a)$$

$$\vec{y} = (a, b)$$

we denote where $k \leftarrow \mathcal{K}$, $\vec{c}_0 = (\text{Enc}_k(a), \text{Enc}_k(a)), \vec{c}_1 = (\text{Enc}_k(a), \text{Enc}_k(b))$.

Let $\mathcal{A} : \mathcal{C} \rightarrow \{0, 1\}$ be defined by $\mathcal{A}(z_1, z_2) = \mathbb{1}[z_1 = z_2]$, then

$$\Pr_k [\mathcal{A}(\vec{c}_0) = 1] = 1 \neq 0 = \Pr_k [\mathcal{A}(\vec{c}_1) = 1]$$

and thus for any $\varepsilon \in [0, 1]$ one can distinguish between the distributions, so the scheme is insecure for multiple messages. ■

2.4 Stream Ciphers

Algorithm 2.1 [From 1-Message Security to Multiple Message Security] Let (Enc, Dec) be secure for single message. Construct a new scheme with multiple messages security:

- Initialization: Let $S \leftarrow \mathcal{K}$ be a fresh random key.
- Encrypt a message m with state S :
 - Choose a new random key $S' \leftarrow \mathcal{K}$, send $\text{Enc}_S(m, S')$.
 - Update the state to S' .
- To decrypt a ciphertext c with state S :
 - Compute $\text{Dec}_S(c) = (m, S')$, output m .
 - Update the new state to S' .

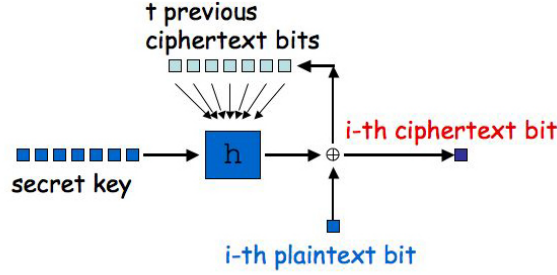
Alternatively, use a **stateful PRG**:

PRG(current-state) outputs (new-state, pseudorandom-block)

Algorithm 2.2 [Stream Cipher (Synchronous)] Stateful PRGs are the basis for **stream ciphers**:

- Let $S_0 \xleftarrow{R} \mathcal{K}$ be the random secret key, used as the initial state.
- To encrypt the i -th bit of the message m_i do:
 - $(b_i, S_{i+1}) \leftarrow \text{PRG}(S_i)$
 - Output $c_i = m_i \oplus b_i$

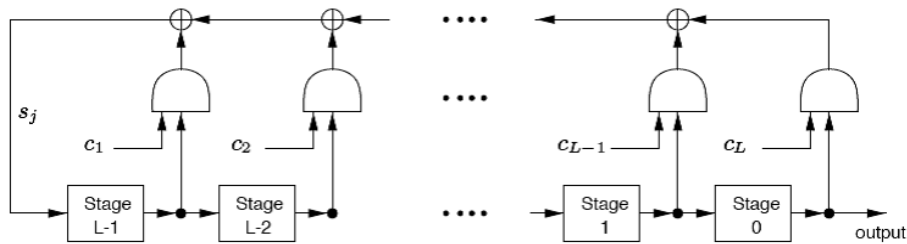
Problem 2.1 What happens if one bit of the transmission is lost ?



Algorithm 2.3 [Asynchronous (Self Synchronizing) Stream Ciphers]

- Start with a secret, random key ("seed"), k .
- Generate online a keying stream.
- The i -th bit is a function of the seed and the recent t bits of the ciphertext, $(c_{i-t}, \dots, c_{i-1}) = \text{public state}$.
- Specifically, encryption is $c_i = m_i \oplus h(k, c_{i-t}, \dots, c_{i-1})$
- While decryption is $m_i = c_i \oplus h(k, c_{i-t}, \dots, c_{i-1})$

2.4.1 Linear Feedback Shift Registers



Definition 2.8 [Linear Feedback Shift Register] An **LFSR** consists of L stages (or delay elements) that store bits. The device's hardwiring is defined by coefficients $c_i \in \{0, 1\}$, with c_L always set to 1.

- **Initialization:** The process begins with an initial state $[s_{L-1}, \dots, s_1, s_0]$.
- **Feedback and Output:** The output sequence $s_0, s_1, s_2, \dots$ is determined by a recursion:

$$s_j = \sum_{i=1}^L c_i s_{j-i} \mod 2$$

Remark 2.4 With a precise choice of wiring and initialization, the resulting stream can achieve a very long period.

2.4.2 RC4 (Ron's Code 4)

Algorithm 2.4 [RC4] Operates on a state consisting of a permutation S of $\mathbb{Z}_{256}$ and two-8bit index-pointers i, j .

Initialization: (Shuffling S)

1. $j = 0$
2. $S_0 = 0, S_1 = 1, \dots, S_{255} = 255$
3. Let the key be $k_0, \dots, k_{255}$ (repeating bits if key has fewer bits)
4. For $i = 0$ to 255
 - $j = (j + S_i + k_i) \bmod 256$
 - Swap S_i and S_j

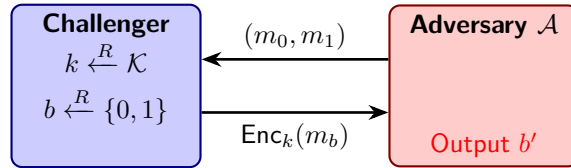
An output byte B is generated as follows:

- $i = i + 1 \bmod 256$
- $j = j + S_i \bmod 256$
- Swap S_i and S_j
- $r = S_i + S_j \bmod 256$
- $B = S_r$

B is the next keying stream byte, and is XORed with the next plaintext byte to produce ciphertext byte.

2.5 Indistinguishability under Chosen Plaintext Attack

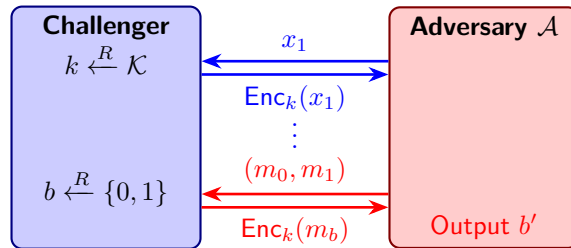
Theorem 2.3 [Ciphertext Indistinguishability: Alternative Definition] Consider the following game:



Then, (Enc, Dec) has (t, ε) -computational secrecy if and only for every t -time adversary $\Pr[b' = b] \leq \frac{1}{2} + \frac{1}{2}\varepsilon$.

Proof was left to HW1, Q2.

Definition 2.9 [Indistinguishability under Chosen Plaintext Attack] Consider the following game:



The game has two phases:

1. $\mathcal{A}$ is allowed to **adaptively choose many encryptions**
2. $\mathcal{A}$ chooses a test $m_0, m_1 \in \mathcal{M}$ and tries to **distinguish** $\text{Enc}_k(m_0)$ from $\text{Enc}_k(m_1)$

We then say, (Enc, Dec) is (t, ε) -CPA secure if for every t -time adversary $\Pr[b' = b] \leq \frac{1}{2} + \varepsilon$.